ACME Mine Tyre, Wheel and Rim Management Procedure

(Dummy document for RAG testing – not for operational use)

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1. Document Control

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Sponsor	Site Senior Executive (SSE)
Approver	Operations Manager
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2. Purpose

To define the minimum requirements for the safe lifecycle management of tyres, wheels and rims at ACME Mine, including selection, inspection, handling, fitting, inflation, operation, repair and disposal, to eliminate or minimise the risk of tyre and wheel related incidents (e.g. explosions, separations, roll-aways and dropped loads).

3. Scope

This procedure applies to all ACME Mine personnel and contractors involved with or potentially exposed to heavy vehicle and light vehicle tyres, wheels and rims on site (surface operations, workshops, tyre bays, laydown areas and haul roads).

4. References

- Recognised Standard 13 – Tyre, Wheel and Rim Management (Qld RHSQ).
- Coal Mining Safety and Health Act 1999 & Regulation 2017.
- ACME Risk Management Standard (ACME-SHMS-RSK-ST-001).
- OEM tyre and wheel technical data (Michelin, Bridgestone, Goodyear, etc.).
- AS/NZS 4457 series – Earth-moving machinery – Off-the-road tyres – Rims and rim components.

5. Definitions & Acronyms

- **CMW** – Coal Mine Worker.
- **SSE** – Site Senior Executive.
- **RS13** – Recognised Standard 13.
- **TWR** – Tyre, Wheel and Rim.
- **OTR Tyre** – Off-The-Road tyre used on mining equipment.
- **Burst Zone/Exclusion Zone** – Area to be barricaded during inflation/deflation.
- **Condition Coding** – Standardised tyre condition codes (Appendix B).
- **WRAC/JSA** – Workplace Risk Assessment and Control / Job Safety Analysis.

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6. Roles and Responsibilities

SSE
- Ensure this procedure is implemented and resourced.
- Authorise deviations via Change Management process.

Maintenance Superintendent (Document Owner)
- Maintain this procedure and associated standards.
- Ensure tyre bay, tooling and equipment are compliant and inspected.

Tyre Bay Supervisor
- Allocate trained personnel for tyre tasks.
- Verify pre-task risk assessments and permits are completed.
- Maintain tyre inventory, condition records and pressure logs.

Authorised Tyre Fitters/Technicians
- Conduct inspections, fitting, inflation and repairs following this procedure.
- Establish and maintain exclusion zones during pressurised work.
- Record all work in TWR Register.

Mobile Equipment Operators
- Complete pre-start tyre checks (Appendix A).
- Report damage, abnormal heat, vibration, or pressure alarms immediately.
- Do not drive on a tyre below minimum pressure or with visible damage.

Stores/Procurement
- Source tyres/wheels/rims meeting OEM and ACME specifications.
- Maintain storage conditions to prevent degradation.

All Contractors
- Comply with this procedure or demonstrate equivalent standards (approved by SSE).

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7. Risk Management Requirements

- A WRAC/JSA must be completed for all non-routine tyre tasks (e.g., roadside tyre change, split rim work).

- Hazards to consider include: stored energy (compressed air), fire, explosion due to pyrolysis, rim separation, dropped loads, pinch points, vehicle movement, heat, chemical exposure (inflation media).
- Controls must follow the hierarchy: eliminate, substitute, engineer, admin, PPE.
- Energy isolation per Section 11 is mandatory before removing tyres/wheels.

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8. Tyre, Wheel & Rim Lifecycle Management

8.1 Selection & Procurement

- Only purchase tyres/wheels/rims approved by Engineering (load rating, speed rating, temperature class).
- Tyres must match OEM rim size and profile.
- Retread and repair vendors must be pre-qualified.

8.2 Transport & Storage

- Tyres transported upright and secured; no stacking > 3 high unless OEM approved.
- Store in covered, cool areas away from hydrocarbons, ozone sources and direct sunlight.
- Rims stored on stands to prevent damage to bead seats; split ring components kept matched.

8.3 Inspection & Condition Coding

- All tyres inspected on receipt using Appendix B coding.
- Operators complete visual checks each shift; Tyre Bay inspects heavy fleet tyres weekly.
- Any tyre with cuts exposing cords, bulges, or below 80% of set pressure must be removed as soon as practicable.
- Rims inspected for cracks, corrosion, elongation of bolt holes; suspect rims quarantined and NDT tested.

8.4 Fitment/Removal

- Only trained personnel using certified bead breakers, manipulators and torque tools.
- Use rim charts and OEM torque specifications (Appendix C).
- Clean and lubricate bead seats with approved compounds only.
- Never mix components from different rim manufacturers unless certified compatible.

8.5 Inflation & Pressure Control

- Inflate in an approved inflation cage or behind a blast wall, with remote clip-on chuck and 3 m hose.
- Establish an exclusion zone equal to 1.5 × tyre diameter in all radial directions, barricaded and signed.
- Stand out of the trajectory path (side-on, never front-on).
- Use calibrated gauges (±2% accuracy).
- Record cold inflation pressures in the TWR Register; adjust for ambient temperature per OEM chart.
- Deflation must occur prior to removing any clamp rings or split components.

8.6 Operation & Monitoring

- Telematics (if fitted) to alert for low/high pressure or temperature; alarms investigated immediately.
- Haul road design maintained to RS19 standards to minimise tyre damage.
- Speeds and loads must not exceed tyre ratings.

8.7 Repair & Retread

- Only qualified vendors conduct hot/cold repairs.
- Repairs beyond OEM limits are prohibited.
- Post-repair tyres re-inspected and pressure tested before service.

8.8 Disposal

- Tyres condemned when tread depth < OEM minimum, irreparable damage, or > 6 years old (unless engineering approval).
- Dispose/recycle via licensed waste contractor; rims scrapped after de-beading and de-energising.

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9. Standard Work Process – Changing a Haul Truck Tyre (Example)

1. **Plan and Risk Assess** – Complete JSA, confirm tools, pressure, weather, location.
2. **Isolate Equipment** – Park on level ground, park brake, wheel chocks, isolation/LOTO applied.
3. **Establish Exclusion Zone** – Barricade, signage, spotter in place.
4. **Deflate Tyre** – Use remote deflation tool; verify zero pressure.
5. **Remove Wheel Nuts** – Use calibrated torque wrench; follow star pattern.
6. **Dismount Wheel/Tyre** – Use certified lifting devices; never work under suspended loads.
7. **Inspect Components** – Rim, bead seats, lock rings, tyre condition.
8. **Fit Replacement** – Clean/lube, assemble as per rim chart, verify component match.
9. **Inflate Safely** – Remote inflation, cage/blast wall, monitor pressure increments.
10. **Re-torque & Re-check** – After first hour of operation and at next service.
11. **Record** – Update TWR Register with tyre ID, rim ID, pressure, torque, fitter’s name.

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10. Exclusion Zones & Barricading

- Minimum exclusion radius = 1.5 × tyre diameter (or OEM specified).
- Use physical barricades (cones/tape insufficient for high-risk jobs), signage “DANGER – TYRE INFLATION”.
- Only essential personnel inside zone; communication via radio/hand signals.

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11. Energy Isolation & Lockout

- Apply ACME Isolation Procedure (ACME-SHMS-ENG-PR-006).
- Isolate: mechanical (jacks/stands), hydraulic (bleed down), electrical (battery isolator), pneumatic (deflate).
- Verify zero energy prior to rim disassembly.

12. Tools & Equipment Requirements

- Inflation cages/barriers certified to withstand tyre explosions.
- Bead breakers, manipulators and torque tools inspected quarterly.
- Pressure gauges calibrated 6-monthly (retain certificates).
- Tyre handlers rated for the maximum tyre weight plus 20% margin.

13. Training & Competency

- Tyre fitters must complete nationally recognised competency: AURKTJ011 (or equivalent).
- Annual refresher training on RS13 changes and site learnings.
- Operators trained in pre-start tyre inspection and reporting.
- Competency records maintained in LMS.

14. Auditing, Verification & Review

- Monthly tyre bay inspection (Supervisor).
- Quarterly internal audit against RS13 elements.
- Annual third-party audit (external SME).
- Non-conformances entered into ACME Action Tracker; corrective actions verified.

15. Records & Data Management

- TWR Register to capture: tyre ID, rim ID, fitment date, machine, position, pressure history, repairs, removal reason, disposal date.
- All JSAs, permits, inspection forms retained for minimum 2 years or as required by legislation.

16. Emergency Response

- If a tyre fire occurs: evacuate to minimum 300 m (light vehicles) / 500 m (haul trucks) upwind, notify Control Room. Do NOT apply water directly—follow OEM guidance.
- Tyre burst injury: Call emergency response, treat as high-energy trauma.
- Rim separation incident: cease all tyre work, secure area, notify SSE and Inspector of Mines as required.

17. Change Management

- Any deviation from this procedure or introduction of new tyre technology must go through ACME Management of Change (MOC) process.

18. Document Review

- Reviewed biennially or after any tyre-related incident, audit finding, or RS13 update.

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19. Appendices

Appendix A – Operator Pre-Start Tyre Checklist (Example)

- Visible cuts, bulges, chunking?
- Embedded objects (rocks, metal)?
- Tyre appears under/over-inflated?
- Hot smell/smoke?
- Wheel nuts present and visually intact?
- Report abnormalities to Dispatch immediately.

Appendix B – Tyre Condition Coding (Example)

Code	Description	Action
C0	New / Serviceable	Monitor
C1	Minor damage (≤ 10 mm)	Monitor / schedule repair
C2	Moderate damage (cords not exposed)	Remove when practical
C3	Severe damage (cords exposed, bulge)	Remove immediately
C4	Scrap / Beyond repair	Dispose / quarantine

Appendix C – Typical Torque Values (Example Only)

- 51" Rim Clamp Nuts – 1,150 N·m
- 33" Wheel Nuts – 750 N·m
  (Verify actual values with OEM manuals)

Appendix D – Tyre Change JSA Template (Abbreviated)

1. Task steps listed.
2. Hazards identified for each step.
3. Controls (engineering, admin, PPE).
4. Residual risk rating.
5. Authorisations and sign-off.

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